

PyLM: A Python implementation for Landscape Mosaic analysis

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Summary

Landscape ecology is the study of how different land uses and natural areas are arranged across a region, and how these spatial patterns affect biodiversity, ecosystem health, and human impacts. To measure and track these patterns, scientists use a range of tools and metrics that capture features such as connectivity, fragmentation, and the balance between natural and developed land. One such method is the Landscape Mosaic (LM) approach, originally developed by Riitters et al. (2009) and further developed by Vogt et al. (2024) which classifies land into categories based on the mix of agriculture, natural habitats, and developed areas, providing an integrated view of how humans influence ecosystems. Until recently, LM was only available through a specialized software package (i.e., GuidosToolbox - Vogt & Riitters (2017)), which limited its flexibility and integration with other tools. To address this, hereafter we present PyLM, a Python-based implementation of the LM model, making it easier for researchers, planners, and conservationists to analyze land cover maps, generate statistics, and embed results into broader environmental workflows. This enhances accessibility, supports sustainability assessments, and strengthens the ability to monitor landscapes over time.

Statement of need

Landscapes are defined as mosaics of different land cover, land use, habitats, and ecosystems [REF]. Landscape ecology is a field of ecology that specifically examines the spatial patterns of ecosystems and the processes that shape them across heterogeneous environments [REF]. It emphasizes the relationships between spatial configuration (e.g., size, shape, distribution) and ecological dynamics including species movement, biodiversity, and ecosystem functioning [REF]. To quantify these spatial patterns, a variety of metrics are available to account for characteristics such as patch area, diversity, density, connectivity and fragmentation [REF Computational methods; PyLandStats, NLMpy]. These metrics serve as essential tools for conservation planning, habitat management, and sustainability research by providing a standardized way to analyze landscape structure, detect changes over time, and assess the impacts of natural disturbances or human activities like urbanization, agriculture, and deforestation. To identify and assess spatial patterns, different methods are available enabling to monitor landscape's evolution and linking them to specific ecological processes [REF]. The Landscape Mosaic (LM) approach provides a framework for quantifying spatial heterogeneity by employing a tri-polar classification scheme based on three land cover classes [REF Riitters]. The method applies a moving window to quantify the relative proportions of the three classes within each focal grid cell. These proportions are subsequently assigned to one of 19 mosaic categories, determined by threshold criteria that capture the presence, dominance, or exclusivity of each class. The LM approach is designed to characterize the spatial juxtaposition of anthropogenic land cover (i.e., developed and agricultural areas) relative to natural land cover. By simultaneously evaluating these three dominant land cover types, the approach captures landscape composition,

connectivity, and heterogeneity, thereby providing an integrated measure of human influence on landscapes. The indicator assigns each location to a category based on the relative proportions of Agriculture, Natural, and Developed land within its surrounding neighborhood. This classification supports the quantification and spatial assessment of landscape capacities to sustainably supply ecosystem services [REF]. The Landscape Mosaic provides a framework for assessing the state, pressures, and temporal dynamics of landscapes, including the intensity of land use across agricultural, urban, and natural ecosystems, as well as their complex interactions under changing environmental and socio-economic conditions. Currently there is only one implementation of the LM framework within the GuidosToolbox (Graphical User Interface for the Description of image Objects and their Shapes - GTB) [REF]. GTB is a free and open-source software with a set of customized and thematically grouped raster image processing routines mostly used in environmental, ecological, forestry, land cover / land use, and landscape-ecology contexts. It is available through a Desktop application, for Windows, Mac and Linux as well as a command-line Linux server application for mass processing (GuidosToolbox Workbench GWB). Despite the fact that the LM model is available as single tool within GTB/GWB, its interaction with other software/models is limited and therefore there is a need for having a version that could be available (1) as a standalone tool, (2) that could be executed both on a desktop or a server, (3) that could be easily embedded in processing chains/workflows or integrated in other analytical framework like LivingEarth [REF], (4) based on common data science language such as Python. Based on these considerations, PyLM provides a Python implementation of the Landscape Mosaic model enabling users to process any land cover map and generate the different stratification layers as well as relevant statistics (e.g., heatmap). PyLM offers increased accessibility, flexibility, and computational power for landscape ecology, allowing for easier analysis, customization, and integration with other open-source tools for research, conservation, and planning. PyLM could be executed either as a standalone Python script, within a Jupyter Notebook environment, and ultimately could be easily embedded in any processing workflow. Overall PyLM simplifies the production of key landscape metrics necessary for environmental monitoring.

Main modules

Linear process (each module the subsequent one) (see for modules description: <https://publications.jrc.ec.europa.eu/repository/handle/JRC120383>) (1) Initialization (2) Input/Output data + map conversion [optional] (3) LM Analysis (3.1) Moving window - 103 classes - 19 classes (3.2) LM Background (3.3) LM Diversity (3.4) LM Agriculture (3.5) LM Natural (3.6) LM Developed (3.7) LM Anthropropic intensity (3.8) Heatmap

Installation

Require Jupyter Notebook/Lab and Python 3.10 (minimal) Copy the notebook in the folder

Author contributions

Gregory Giuliani authored the original version of the package, developed the pipeline and all of its functions, maintained the package, wrote the documentation, debugged the code, and contributed to the drafting of the article.

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